Deutsch–Jozsa Algorithm Tutorial

Problem Setup

Given a function

$$f : \{0,1\}^n \to \{0,1\},$$

determine whether f is *constant* or *balanced*.

- **Constant:** $f(x)$ is the same for every x.
- **Balanced:** f returns 0 for half the domain and 1 for the other half.
- We are given an *oracle*: a black box that evaluates f at x.

Classically, the worst case would be to run the oracle

$$2^{n-1} + 1$$

times, one more than half the domain.

Quantumly, we only have to run the oracle once.

We are given a quantum oracle, which is a matrix U, that does the following:

$$U |x\rangle = (-1)^{f(x)} |x\rangle .$$

This oracle does not write $f(x)$ into a new output bit. Instead, it places $f(x)$ into the *phase* of $|x\rangle$.

Step 0: Initialize the Quantum Circuit

Since we are working with length n bitstrings, we need n qubits.

Qubits are always initialized to 0. Thus, our initial state is

$$|\psi_0\rangle = |0^n\rangle .$$

0 ———

2 ———

Step 1: Prepare a Superposition of All Bitstrings

The quantum Hadamard gate, which mathematically is a Hadamard matrix, famously takes $|0\rangle$ and turns it into an equal superposition:

$$H |0\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) .$$

For longer bitstrings, such as $|00\rangle$, applying a Hadamard gate on both qubits creates an even superposition of all possible bitstrings:

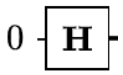
$$H^{\otimes 2} |00\rangle = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle) .$$

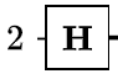
For an arbitrary bitstring x, applying a Hadamard on each qubit results in the following state:

$$H^{\otimes n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} (-1)^{x \cdot z} |z\rangle .$$

Thus,

$$H^{\otimes n} |\psi_0\rangle = H^{\otimes n} |0^n\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle = |\psi_1\rangle .$$





Step 2: Apply the Quantum Oracle U

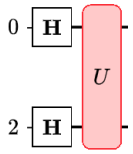
The phase of a quantum state $c\,|x\rangle$, where

$$c = Re^{i\theta} \in \mathbb{C},$$

is $e^{i\theta}$. For example, the phase of $|0\rangle$ is $+1$.

The quantum oracle adds a phase to each bitstring in the superposition:

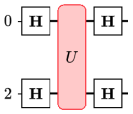
$$\begin{aligned} U\,|\psi_1\rangle &= U\left(\frac{1}{\sqrt{2^n}}\sum_{x\in\{0,1\}^n}|x\rangle\right) \\ &= \frac{1}{\sqrt{2^n}}\sum_{x\in\{0,1\}^n} U\,|x\rangle \\ &= \frac{1}{\sqrt{2^n}}\sum_{x\in\{0,1\}^n}(-1)^{f(x)}\,|x\rangle = |\psi_2\rangle\,. \end{aligned}$$



Step 3: Apply the Hadamard Gate on All Qubits

This is the interference step. Applying the Hadamard gate on all qubits will cause the phases on each state to add/subtract in a way that gives us our final answer.

$$\begin{aligned} H^{\otimes n}\,|\psi_2\rangle &= H^{\otimes n}\left(\frac{1}{\sqrt{2^n}}\sum_{x\in\{0,1\}^n}(-1)^{f(x)}\,|x\rangle\right) \\ &= \frac{1}{\sqrt{2^n}}\sum_{x\in\{0,1\}^n}(-1)^{f(x)}H^{\otimes n}\,|x\rangle \\ &= \frac{1}{\sqrt{2^n}}\sum_{x\in\{0,1\}^n}(-1)^{f(x)}\left(\frac{1}{\sqrt{2^n}}\sum_{z\in\{0,1\}^n}(-1)^{x\cdot z}\,|z\rangle\right) \\ &= \frac{1}{2^n}\sum_{z\in\{0,1\}^n}\left(\sum_{x\in\{0,1\}^n}(-1)^{f(x)+x\cdot z}\right)|z\rangle\,. \end{aligned}$$



Step 4: Interpret the Result

Inspect the coefficient in front of $|0^n\rangle$. Setting $z = 0^n$, we get

$$x \cdot 0^n = 0,$$

so the amplitude of $|0^n\rangle$ is

$$\frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x)+x\cdot 0^n} = \frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x)}.$$

If f is constant, then

$$\sum_{x \in \{0,1\}^n} (-1)^{f(x)} = \pm 2^n,$$

so the amplitude of $|0^n\rangle$ is

$$\pm 1.$$

Thus, we have 100% chance of measuring all the qubits to be 0.

If f is balanced, then

$$\sum_{x \in \{0,1\}^n} (-1)^{f(x)} = 0,$$

so the coefficient in front of $|0^n\rangle$ is 0, meaning we have a 0% chance of measuring all zeros.

Thus, we measure all of our qubits.

$$\boxed{\text{If we measure } 0^n, \text{ then } f \text{ is constant.}}$$

$$\boxed{\text{Otherwise, } f \text{ is balanced.}}$$